

Supply, Demand, and Government Policies

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 6** — Supply, Demand, and Government Policies.

Read the chapter alongside these slides.

Where we are going

1. **Price ceilings** — a legal maximum, and what it leaves behind.
2. **Price floors** — the mirror image.
3. **The minimum wage** — the most argued-about floor in economics.
4. **What prices were doing all along** — why economists flinch at controls.
5. **A tax on sellers** — the wedge between two prices.
6. **A tax on buyers** — the same tax, collected at the other end.
7. **The two designs compared** — and why the law's choice does not matter.
8. **When the split is not even** — change one slope and watch.
9. **Revenue, and the missing trades** — what the government collects, and what nobody collects.
10. **Who was actually taxed** — a tax on yachts that landed on boatbuilders.

PART 1

Price Ceilings

The government enters the market

Our coffee market:

$$Q^d = 70 - 10P, \quad Q^s = 10P - 10 \quad \Rightarrow \quad P^* = \$4, \quad Q^* = 30$$

(thousands of cups per week).

Today the government reaches into it, with the only two instruments it has:

- **Price controls** — a limit the price may not cross.
- **Taxes** — a charge on every unit traded.

Both have effects their architects did not intend. We will compute them.

When government sets the price

So far the price was free to move. Now suppose it is not.

Price ceiling. A legal *maximum* on the price at which a good can be sold.

Rent control is the standard example. The motive is usually to protect buyers.

Whether it does anything at all depends on where it is set.

Two cases, and only one of them matters

Not binding. Ceiling *above* equilibrium. The market was already below it, so nothing changes.

Our market clears at \$4. A ceiling of \$5 does nothing at all.

Binding. Ceiling *below* equilibrium. The market cannot reach \$4, so the ceiling price rules.

Always ask first: is it above or below equilibrium?

A binding ceiling at \$3

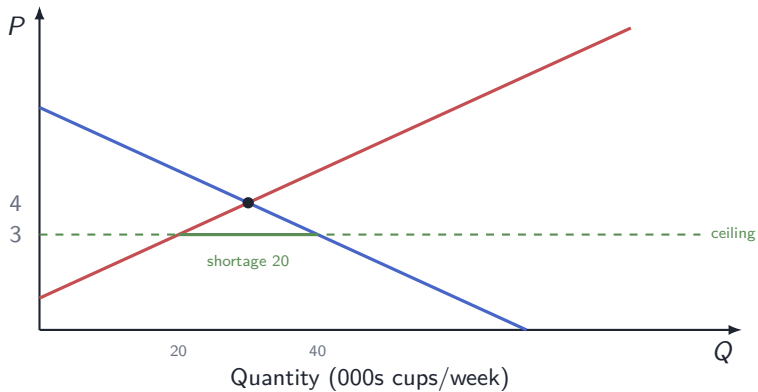
$$Q^d = 70 - 10(3) = 40, \quad Q^s = 10(3) - 10 = 20$$

Shortage of 20 thousand cups per week.

And it **cannot** self-correct. Normally the shortage would push the price up to \$4 — but that is now illegal.

The shortage is permanent for as long as the ceiling stands.

A binding ceiling, drawn



What actually happens under a shortage

Only 20 thousand cups exist. Forty thousand are wanted. **Something has to ration them.**

Price is no longer allowed to do it, so other mechanisms take over:

- Queues — rationing by time rather than money
- Sellers favoring regular customers
- Quality falling, because sellers can sell anything
- Black markets

Buyers who do get coffee pay less. Buyers who do not, get nothing.

Give it time: the shortage grows

Elasticity rises with the time horizon — buyers and sellers need time to find alternatives. Double both slopes through the same equilibrium, and rerun the \$3 ceiling:

$$\text{long run: } Q^d = 110 - 20P, \quad Q^s = 20P - 50$$

	Short run	Long run
Q^d at \$3	$70 - 10(3) = 40$	$110 - 20(3) = 50$
Q^s at \$3	$10(3) - 10 = 20$	$20(3) - 50 = 10$
Shortage	20	40

A ceiling looks mildest in exactly the period when it is judged.

Active Learning 1

Our market: $Q^d = 70 - 10P$, $Q^s = 10P - 10$, $P^* = 4$.

1. A ceiling is set at \$6. Is it binding? What happens?
2. A ceiling is set at \$2. Is it binding? Compute the shortage.
3. Which of the two would a politician rather announce, and which does more?

Four minutes. Question 3 has no arithmetic.

PART 2

Price Floors

The mirror image

Price floor. A legal *minimum* on the price at which a good can be sold.

The minimum wage is the standard example. The motive is usually to protect *sellers*.

Not binding — floor *below* equilibrium. Nothing happens.

Binding — floor *above* equilibrium. The floor price rules.

Everything reverses, including the direction of the test.

A binding floor at \$5

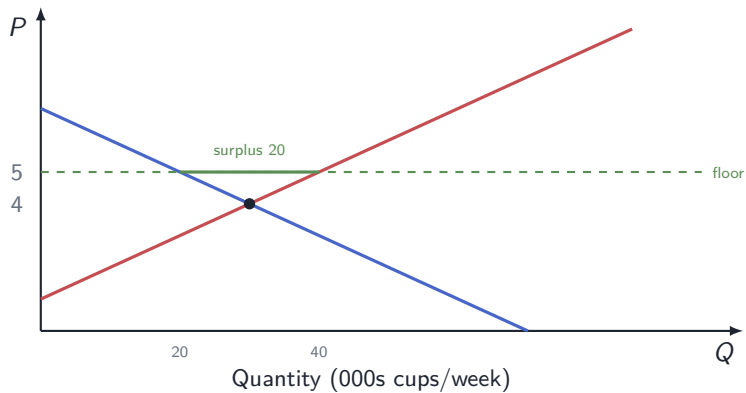
$$Q^d = 70 - 10(5) = 20, \quad Q^s = 10(5) - 10 = 40$$

Surplus of 20 thousand cups per week.

Normally the surplus would push the price down to \$4. That is now illegal.

The surplus is permanent for as long as the floor stands.

A binding floor, drawn



Who gains and who loses

Sellers who sell at the floor price do better — \$5 instead of \$4.

Sellers who cannot find a buyer sell nothing at all. Only 20 of the 40 on offer are bought.

A binding floor helps the sellers who trade and hurts those who cannot.

That is the shape of the minimum wage debate: workers who keep their jobs earn more; the argument is about how many do not.

One test, both instruments

	Price ceiling	Price floor
Legal...	maximum	minimum
Binds when...	<i>below P^*</i>	<i>above P^*</i>
Creates...	shortage	surplus
Rationed by...	queues, favoritism	sellers left unsold
Standard example	rent control	minimum wage

Binding is about position relative to equilibrium — never about how large the number is.

Active Learning 2

Our market again: $P^* = 4$, $Q^* = 30$.

1. A floor is set at \$3. Binding? What happens?
2. A floor is set at \$6. Binding? Compute the surplus.
3. Both a \$3 *ceiling* and a \$6 *floor* are imposed at once. What happens?

Four minutes. Question 3 is harder than it looks.

PART 3

The Minimum Wage

The wage is a price

A labor market is a market. The roles are reversed: **firms are the buyers, workers are the sellers.**

A city's market for teenage labor (W in \$/hour, L in thousands of workers):

$$L^d = 150 - 10W \qquad L^s = 10W - 10$$

L^d : the higher the wage, the fewer workers firms hire.

L^s : the higher the wage, the more teenagers look for work.

A minimum wage is a price floor in this market.

Plug the numbers in

W	$L^d = 150 - 10W$	$L^s = 10W - 10$
\$6	$150 - 10(6) = 90$	$10(6) - 10 = 50$
\$7	$150 - 10(7) = 80$	$10(7) - 10 = 60$
\$8	$150 - 10(8) = \mathbf{70}$	$10(8) - 10 = \mathbf{70}$
\$9	$150 - 10(9) = 60$	$10(9) - 10 = 80$
\$10	$150 - 10(10) = 50$	$10(10) - 10 = 90$

Algebra agrees: $150 - 10W = 10W - 10 \Rightarrow 160 = 20W \Rightarrow \mathbf{W^* = \$8, L^* = 70}$.

A floor at \$10

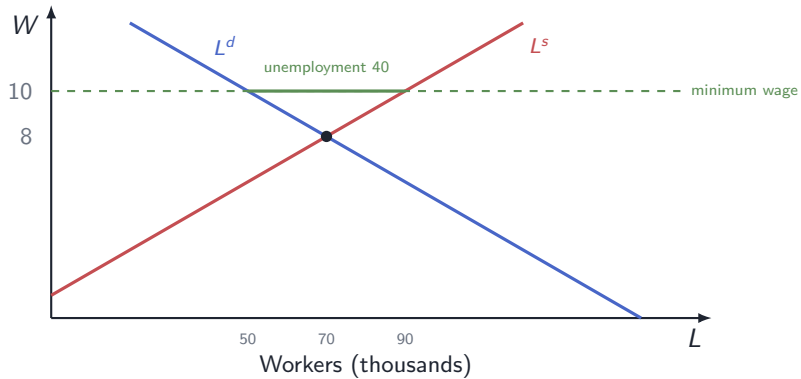
The city sets a minimum wage of \$10. It is *above* $W^* = \$8$, so it **binds**.

$$L^d = 150 - 10(10) = 50, \quad L^s = 10(10) - 10 = 90$$

Surplus of 40 thousand workers — and a surplus of labor has a name: unemployment.

Decompose it: employment fell $70 \rightarrow 50$ (**20 jobs lost**), and the higher wage drew $90 - 70 = \mathbf{20 \text{ more}}$ into the search.

The floor, drawn



Who gains, who loses

Gain: the 50 thousand who keep jobs earn \$10 instead of \$8 — \$2/hour more.

Lose: 20 thousand whose jobs disappear — typically the least skilled, for whom the first job is the training.

Queue: 20 thousand more drawn into a search that cannot succeed.

The floor transfers income toward some low-wage workers and takes the job itself from others.

What the evidence says

The short-run estimates most studies find: **a 10% rise in the minimum wage lowers teenage employment by roughly 1–3%.**

But it is genuinely contested: Card and Krueger (1994) compared fast-food jobs across the New Jersey–Pennsylvania border after New Jersey's 1992 increase and **found no employment fall.**

Long-run effects — training forgone, dropout decisions — are larger and harder to measure.

The theory fixes the direction; the data argue about the size.

Card, D. and A. B. Krueger (1994), “Minimum Wages and Employment,” *American Economic Review* 84(4).

Poorly aimed, in both directions

If the goal is to help **poor families**:

- Many minimum-wage earners are **not in poor households** — teenagers and second earners in middle-income families.
- Many poor households have **no worker at all** — a wage floor cannot reach them.

A targeted alternative exists: the **earned income tax credit** — a wage subsidy paid through the tax system, aimed by household income.

Next part: what every control, however well aimed, does to the price itself.

Active Learning 3

The teen labor market: $L^d = 150 - 10W$, $L^s = 10W - 10$, $W^* = 8$.

1. A minimum wage of \$7 is proposed. Binding? What happens?
2. A minimum wage of \$9 is imposed. Compute unemployment, and split it: jobs lost vs. newly drawn-in seekers.
3. A town raises its minimum wage; a year later teen employment is *higher*. A classmate says the model is refuted. Name one thing (other than the wage) that could have changed instead — and what you would need to check before concluding anything.

Five minutes. The reasoning in 3 is the marks.

PART 4

What Prices Were Doing All Along

The \$4 was doing work

Nobody set the price of coffee at \$4. It **emerged** — two schedules, what buyers will pay and what sellers will accept, meeting.

That number carries, in one figure, information scattered across thousands of people and held in full by nobody:

- how much buyers want coffee, relative to everything else;
- how costly it is to produce.

A market price is not just a cost. It is a message.

The message at work

A frost in Brazil cut the coffee harvest. Supply fell, and the price rose

\$4 $\rightarrow$ \$5.

At \$5: every drinker in the world was told *use less* — and every grower was told *produce more* —

without anyone being informed that a frost had happened at all.

No planner collected the information. The price **moved it**.

A control silences the signal

Now freeze the price at \$3 by law.

The scarcity has not changed. Only 20 thousand cups exist; 40 thousand are wanted.

But the number that carried the news is **frozen**: buyers see cheap coffee and ask for more; sellers see a low price and produce less — each told the *opposite* of the truth.

Queues and favoritism now do the rationing — and they **transmit nothing** to anyone who could ease the scarcity.

The control changes the message, not the scarcity.

Extensions — Hayek and the knowledge problem

Everything on this frame is beyond the department's required material.

Hayek's point: the knowledge a planner would need — who wants what, where, at what cost, this week — **never exists in one place**. It is dispersed, local, and constantly changing.

The price system is what moves it: “a system of telecommunications.” Nobody needs to know *why* the number changed to respond correctly.

A binding control is, on this view, a broken wire: **the scarcity persists; the news of it stops traveling.**

Hayek, F. A. (1945), “The Use of Knowledge in Society,” *American Economic Review* 35(4), 519–530.

Helping without silencing

The motive — affordable housing, a living wage — is not the mistake. **The instrument is.**

Policies that *shift the curves* leave the price free to speak:

- **Rent subsidies** — shift housing demand; builders still hear “build.”
- **Wage subsidies (EITC)** — raise take-home pay without pricing anyone out of a job.
- **Build more** — shift supply itself.

Cap the price and you fight the message. Shift the curves and you change the market it describes.

Active Learning 4

A hurricane knocks out a region's ports. Within days, the price of plywood **triples**. The state imposes a cap at the old price.

1. What was the tripled price *saying*, and to whom?
2. Under the cap, how does the existing plywood get allocated? Who is more likely to get it — the family with the wrecked roof, or the neighbor who arrived at the store first?
3. Does the cap change how much plywood exists in the region in week one? What does it change about week four?

Four minutes. No arithmetic — the argument is the answer.

PART 5

A Tax on Sellers

The second instrument

The city levies a **\$1 tax on every cup sold, collected from the seller.**

Who bears a tax is already settled: the more inelastic side bears more, because the side with fewer alternatives cannot escape the tax by walking away. That half is done and stays done.

Today, what the tax does **to the market itself**:

- the **wedge** it drives between two prices,
- the **revenue** it raises,
- the **trades it prevents**.

Same tools as always: write it, solve it, draw it.

Two ways to write the same curve

Every equation in this lecture has Q as the subject. Every diagram has Q on the horizontal axis. So each curve has to be turned round before it can be plotted.

	Solved for Q	Solved for P
Demand	$Q^d = 70 - 10P$	$P = 7 - Q/10$
Supply	$Q^s = 10P - 10$	$P = 1 + Q/10$

Check at $Q = 30$: $7 - 3 = 4$ and $1 + 3 = 4$. ✓

And the tax is about to move an intercept: $Q^s = 10P - 20$ inverts to $P = 2 + Q/10$.
The 1 became a 2 — up by exactly the tax.

Write the equation

Buyers pay the till price P . Sellers hand over \$1, so they **keep** $P - 1$.

Sellers decide how much to produce by what they *keep*:

$$Q^s = 10(P - 1) - 10 = 10P - 20$$

At every till price, sellers behave as if the price were \$1 lower — **the supply curve shifts up by exactly the tax.**

Demand is untouched: $Q^d = 70 - 10P$. Buyers respond to the till price, and nothing changed for them — *yet*.

Plug the numbers in

P	$Q^d = 70 - 10P$	$Q^s = 10P - 20$
\$4.00	$70 - 10(4.00) = 30$	$10(4.00) - 20 = 20$
\$4.25	$70 - 10(4.25) = 27.5$	$10(4.25) - 20 = 22.5$
\$4.50	$70 - 10(4.50) = \mathbf{25}$	$10(4.50) - 20 = \mathbf{25}$
\$4.75	$70 - 10(4.75) = 22.5$	$10(4.75) - 20 = 27.5$

At the old price \$4 there is now a **shortage** (30 wanted, 20 supplied): the price must rise. The rows cross at **\$4.50**.

Solve it, and check both ways

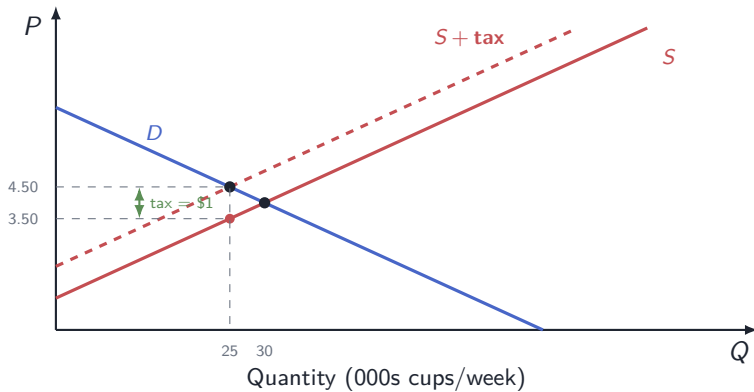
$$70 - 10P = 10P - 20 \Rightarrow 90 = 20P \Rightarrow P = \$4.50$$

Buyers pay (till price) P_B	\$4.50
Sellers keep $P_S = 4.50 - 1.00$	\$3.50
Quantity	25

Check by both routes: demand $70 - 10(4.50) = 25$; supply $10(3.50) - 10 = 25$. ✓

Same number, two roads — if they disagreed, we would know to go looking.

Draw the graph



The wedge

Tax wedge. A tax separates the price buyers pay from the price sellers keep. The gap between them is always exactly the tax: $P_B - P_S = T$.

Here: $4.50 - 3.50 = \$1.00$ ✓

There is no longer *one* price. Buyers trade along D at \$4.50; sellers along S at \$3.50; the government stands in the gap.

And the quantity fell, $30 \rightarrow 25$. Hold that — it is Part 9.

The split — and why it is even here

Buyers: $4.50 - 4.00$	$+\$0.50$
Sellers: $4.00 - 3.50$	$-\$0.50$
Check: $0.50 + 0.50$	$= \$1.00 \checkmark$

Even split, because the elasticities are **equal** at equilibrium:

$$|E_d| = 10 \times \frac{4}{30} = 1.33 \quad E_s = 10 \times \frac{4}{30} = 1.33$$

The governing rule: the more inelastic side bears more. This market is deliberately symmetric, so the machinery — not the split — gets your attention.

Part 8 changes one slope, and the split stops being even.

Before and after, in one table

	No tax	\$1 tax on sellers
Buyers pay	\$4.00	\$4.50
Sellers keep	\$4.00	\$3.50
Wedge	—	\$1.00
Quantity	30	25

One price became two, and five thousand weekly trades disappeared.

Active Learning 5

Same market: $Q^d = 70 - 10P$, $Q^s = 10P - 10$. The city **doubles** the tax to \$2 per cup, still collected from sellers.

1. **Predict before solving:** the wedge, and how the burden will split. Say *why*.
2. Write the new supply equation.
3. Solve: P_B , P_S , and Q . **Check: the two burdens must sum to \$2.00.**

Five minutes. Prediction committed before part 3.

PART 6

A Tax on Buyers

Move the collection point

Same \$1 tax — but now **collected from buyers** at the register, on top of the till price.

Buyers care about their **all-in cost** $P + 1$:

$$Q^d = 70 - 10(P + 1) = 60 - 10P$$

The demand curve shifts down by exactly the tax. Supply is back to its original:
 $Q^s = 10P - 10$.

Politicians distinguish sharply between the two designs. Let us see if the market does.

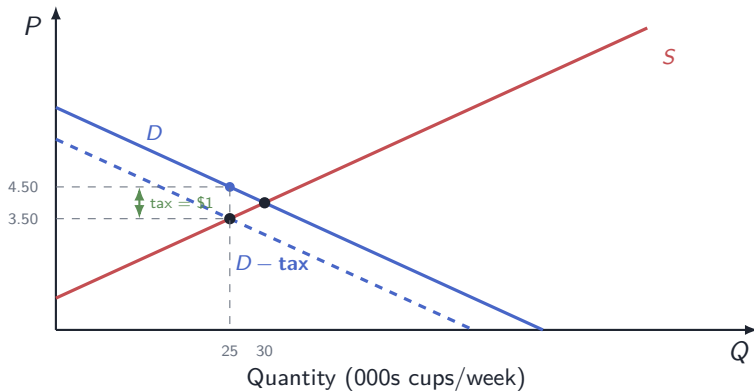
Solve it, and check both ways

$$60 - 10P = 10P - 10 \Rightarrow 70 = 20P \Rightarrow P = \$3.50$$

Till price	\$3.50
Buyers pay all-in $P_B = 3.50 + 1.00$	\$4.50
Sellers keep P_S	\$3.50
Quantity	25

Check both ways: demand $60 - 10(3.50) = 25$; supply $10(3.50) - 10 = 25 \checkmark$

Same wedge, drawn



Active Learning 6

A market for car washes: $Q^d = 96 - 8P$, $Q^s = 4P - 12$. Without a tax the price is \$9 and 24 thousand washes a month trade.

The city puts a **\$3 tax on every wash, collected from customers** at the till.

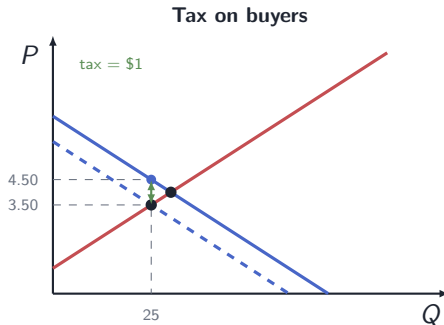
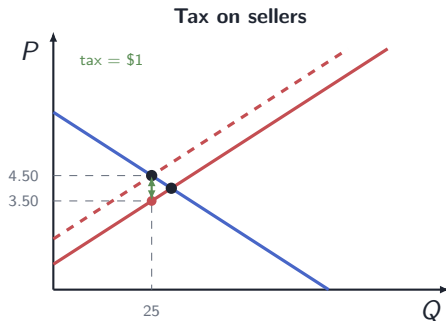
1. Write the new demand equation. Which way does the curve move, and by how much?
2. Solve for the till price, the all-in price buyers pay, and the quantity.
3. **Check:** run the answer through both schedules, and confirm the two burdens sum to \$3.

Five minutes. Watch which price enters which equation.

PART 7

The Two Designs Compared

The same market, twice



The dashed curve is on a different side. Every number is the same.

Line them up

	Tax on sellers	Tax on buyers
Buyers pay all-in	\$4.50	\$4.50
Sellers keep	\$3.50	\$3.50
Quantity	25	25
Wedge	\$1.00	\$1.00
Till price	\$4.50	\$3.50

Only the **till price** — a label — differs.

The legal assignment of a tax has no economic effect.

Who pays, and who bears

Statutory incidence. Who the law requires to remit the tax.

Economic incidence. Whose real income falls.

Who writes the check is law. Who bears the cost is elasticity. The first does not constrain the second.

In this market, from the table you have just read:

- statutory incidence **differs** — sellers remit under one design, buyers under the other;
- economic incidence is **identical** — 50 cents each, both times.

The law splits it anyway: FICA

The US Social Security payroll tax is **split by statute**: 6.2% paid by the employer, 6.2% by the employee.

The equivalence result says: that 50/50 legal split fixes **nothing** about who actually bears it.

Labor *supply* is far less elastic than labor *demand* — most people work regardless. The incidence rule then says: **workers bear most of both halves**, through lower wages.

The statute divides the check-writing. The market divides the burden.

Active Learning 7

A food-truck lunch market: $Q^d = 60 - 2P$, $Q^s = 8P - 40$. A \$2 tax **on the trucks** gives $P_B = 11.60$, $P_S = 9.60$, $Q = 36.8$.

The city now collects the same \$2 **from customers** instead.

1. **Predict all three numbers without solving.** State the principle that entitles you to the prediction.
2. Now confirm: write the new demand equation and solve.

Four minutes. The principle, named, is half the marks.

PART 8

When the Split Is Not Even

One line made flatter

Same market, same demand $Q^d = 70 - 10P$. One change: supply now responds twice as strongly to price, which draws it *flatter*.

$$Q^s = 20P - 50 \quad \text{inverted:} \quad P = 2.5 + Q/20$$

At $P = \$4$: $20(4) - 50 = 30$. **The same equilibrium** — \$4 and 30 thousand cups.
Only the slope of one line has changed.

$$|E_d| = 10 \times \frac{4}{30} = 1.33 \quad E_s = 20 \times \frac{4}{30} = 2.67$$

Both are measured at the same P and Q , so P/Q cancels: the ratio is just 20 : 10.
Supply is twice as elastic, and no solving was needed.

The same tax, a different market

A **\$1.50 tax on sellers**. Sellers keep $P - 1.50$:

$$Q^S = 20(P - 1.50) - 50 = 20P - 80$$

$$70 - 10P = 20P - 80 \Rightarrow 150 = 30P \Rightarrow P = \$5.00$$

Buyers pay P_B	\$5.00
Sellers keep $P_S = 5.00 - 1.50$	\$3.50
Quantity	20

Check both ways: demand $70 - 10(5.00) = 20$; supply $20(3.50) - 50 = 20$. ✓

The burden, and the rule earned

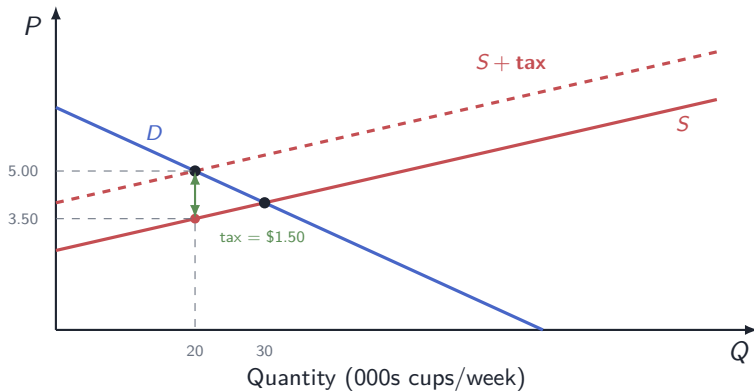
Buyers: $5.00 - 4.00$	$+\$1.00$
Sellers: $4.00 - 3.50$	$-\$0.50$
Check: $1.00 + 0.50$	$= \$1.50 \checkmark$

Buyers bear **twice** what sellers bear. And the elasticity ratio was

$$\frac{E_s}{|E_d|} = \frac{2.67}{1.33} = 2$$

The more inelastic side bears more — here that is demand. Demonstrated, not asserted.

Drawn



Which result depends on what

	Symmetric market	Flatter supply
Supply coefficient (Q per \$)	10	20
$E_s/ E_d $	1	2
Buyers' share of the tax	one half	two thirds
Sellers' share of the tax	one half	one third
Tax on buyers instead	identical	identical

Equivalence is untouched. Here too: a \$1.50 tax on *buyers* gives $Q^d = 55 - 10P$, a till price of \$3.50, an all-in cost of \$5.00 and 20 thousand cups. Identical.

The collection point never matters. The split always does.

Active Learning 8

A third version of the same market: $Q^d = 70 - 10P$, and supply $Q^s = 5P + 10$ — flatter still, and through the same equilibrium (\$4, 30).

1. Give $E_s/|E_d|$ **from the coefficients alone**. Which side bears more now?
2. A **\$1.50 tax on sellers**. Predict the split *before* solving.
3. Solve for P_B , P_S and Q . **Check: the two burdens sum to \$1.50.**

Six minutes. The ratio should take ten seconds.

PART 9

Revenue, and the Missing Trades

What the government collects

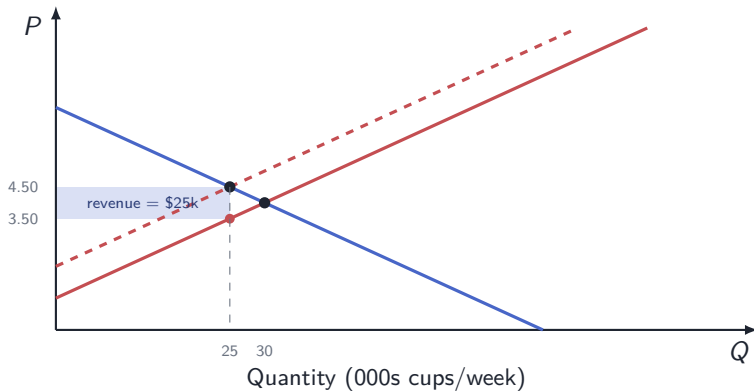
\$1 per cup, on every cup *actually traded*:

$$\text{Revenue} = T \times Q = \$1 \times 25,000 = \$25,000 \text{ per week}$$

The tempting error: $\$1 \times 30,000 = \$30,000$ — taxing the market that **no longer exists**.

The tax shrinks its own base. Every revenue forecast that forgets this comes in high.

The revenue rectangle



Five thousand cups are missing

Q fell 30 $\rightarrow$ 25. Who were those trades? For any cup Q (in 000s): buyer's value $= \frac{70-Q}{10}$, seller's cost $= \frac{Q+10}{10}$.

Cup	Buyer would pay	Seller's cost	Gain from trade	Survives \$1 wedge?
24	$\frac{70-24}{10} = 4.60$	$\frac{24+10}{10} = 3.40$	1.20	yes
26	$\frac{70-26}{10} = 4.40$	$\frac{26+10}{10} = 3.60$	0.80	no
28	$\frac{70-28}{10} = 4.20$	$\frac{28+10}{10} = 3.80$	0.40	no
30	$\frac{70-30}{10} = 4.00$	$\frac{30+10}{10} = 4.00$	0.00	no

Every trade whose gain is smaller than the wedge dies — even though both sides still wanted it.

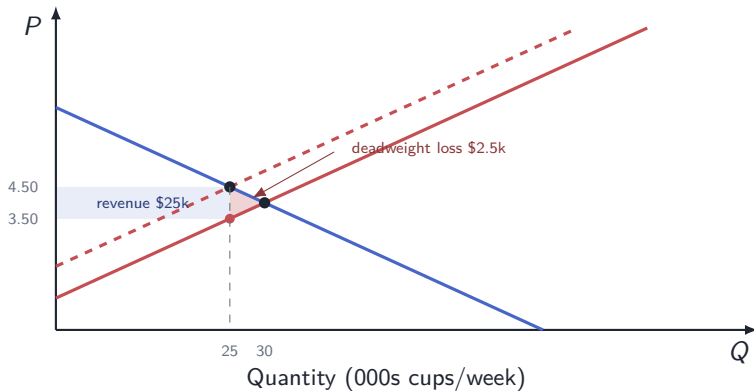
Deadweight loss. The combined value of the trades an intervention prevents — gains from trade that vanish, lost to buyers and sellers and *collected by no one*.

The lost gains shrink steadily from \$1.00 (cup 25) to \$0.00 (cup 30) — a triangle:

$$\text{DWL} = \frac{1}{2} \times \underbrace{5}_{\text{lost trades}} \times \underbrace{\$1}_{\text{wedge}} = \$2,500 \text{ per week}$$

Revenue is a transfer — it leaves the market but arrives at the treasury. Deadweight loss arrives nowhere.

The triangle, drawn



Where this triangle goes next

The ledger, per week:

Government gains (revenue)	+\$25,000
Buyers and sellers lose, combined	−\$27,500
Net loss to society (= DWL)	−\$2,500

Later on, the losses get a formal language (surplus) — and monopoly is measured against competition with *exactly this triangle*.

Deadweight loss is the standard measure of “the market is not doing all it could.”

The other half, recalled

The incidence half, restated: *the more inelastic side bears more of the tax*. A worked pizza market split a \$1 tax 80/20 onto its inelastic side — then flipped the split when the elasticities flipped.

Its loose end: Q fell 70 $\rightarrow$ 66 (hundreds/day) in *both* configurations. You can now name those 4 hundred pizzas/day:

$$\text{missing trades} \quad \text{DWL} = \frac{1}{2} \times 4 \times \$1 = \$200/\text{day}$$

Who pays + what it costs the market = the complete tax analysis.

Active Learning 9

From ALS 5: the \$2 tax gave $P_B = 5$, $P_S = 3$, $Q = 20$ (the \$1 tax gave $Q = 25$, revenue \$25k, DWL \$2.5k).

1. **Predict:** the tax doubled. Does revenue double? Does the deadweight loss double? Commit, with a reason.
2. Compute revenue at \$2.
3. Compute the deadweight loss at \$2. Compare both answers with the \$1 case.

Five minutes. The comparison is the point.

PART 10

Who Was Actually Taxed

1990: a tax on luxury

Congress, seeking revenue from those most able to pay, enacted a **luxury tax**, effective 1991: 10% on yachts above \$100,000 (also jewelry, furs, private aircraft, high-end cars).

The design assumption: *rich buyers will pay it.*

You now own every tool needed to check that assumption before passing the law.

The yacht market, stylized

(P in \$000s per boat, Q boats per month — illustrative numbers, real logic):

$$Q^d = 220 - 8P \qquad Q^s = 2P + 20 \qquad \Rightarrow \qquad P^* = 20, Q^* = 60$$

Elasticities at equilibrium (slope method):

$$|E_d| = 8 \times \frac{20}{60} = \mathbf{2.67} \qquad E_s = 2 \times \frac{20}{60} = \mathbf{0.67}$$

Demand elastic: a yacht is postponable, buyable used, buyable abroad. **Supply inelastic:** boatyards and their skilled workers cannot quickly become anything else.

Predict, using the incidence rule: who bears this tax?

Impose the tax: \$5,000 per boat

Tax collected from sellers: $Q^s = 2(P - 5) + 20 = 2P + 10$.

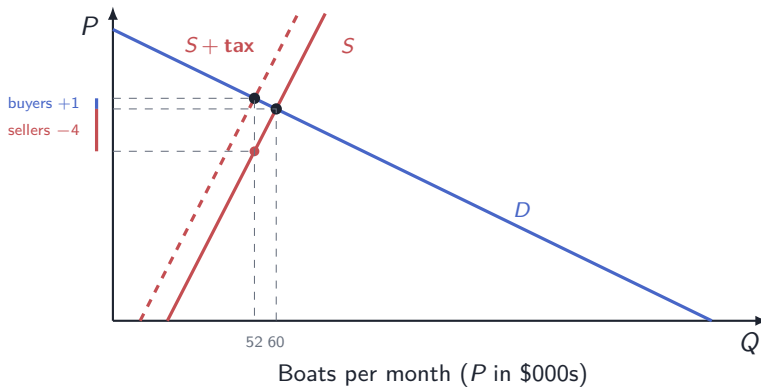
$$220 - 8P = 2P + 10 \Rightarrow 210 = 10P \Rightarrow P_B = 21, \quad P_S = 16, \quad Q = 52$$

Buyers pay +\$1k per boat	20% of the tax
Sellers keep -\$4k per boat	80% of the tax

Check: $1 + 4 = 5 \checkmark$

The “tax on the rich” landed 80% on the boatbuilders.

Drawn: the burden picture



What Congress got

In the stylized market: revenue = $5 \times 52 = \$260\text{k/month}$ — the naive forecast was $5 \times 60 = \$300\text{k}$. Missing trades: 8 boats/month; DWL = $\frac{1}{2} \times 8 \times 5 = \20k/month .

In reality: yacht sales collapsed, boatbuilding towns took layoffs, and revenue fell far short of forecasts.

Congress repealed the boat tax in 1993.

Elastic demand is the worst possible tax base: little revenue, large deadweight loss, and a burden that lands wherever elasticity is lowest — here, on workers.

How to aim a tax

The complete checklist, assembled across two sessions:

1. **Wedge:** $P_B - P_S = T$, whoever writes the check.
2. **Equivalence:** the statutory side is a label.
3. **Incidence:** the more inelastic side bears more.
4. **Revenue:** $T \times Q_{\text{new}}$ — the base shrinks.
5. **Deadweight loss:** $\frac{1}{2} \times \Delta Q \times T$ — grows faster than the tax.

Revenue with least destruction $\Rightarrow$ tax **inelastic** bases. But those are the bases buyers *cannot escape* — which is a fairness question, and **positive analysis cannot settle it**.

Active Learning 10

A state proposes a 5% grocery tax, “**collected from supermarkets, so that families are protected.**” Grocery demand is very inelastic.

1. Does collecting it from the stores protect families? Name the principle.
2. Who bears most of this tax — and is the burden heavier for poor or rich households? Why?
3. Will revenue be large or small? Will the deadweight loss? Justify from elasticity.
4. “It is an efficient tax, so it is a good tax.” Which parts of that sentence can your analysis certify, and which can it not?

Five minutes. Reasoning is the marks — there is almost no arithmetic.

A market price is doing work.

Cap it, and the scarcity stays while the signal dies.

Tax it, and the price splits in two — revenue for the treasury,
and a triangle of trades that never happen, collected by no one.